

Section Activity 5: Turning Up the Heat & Donating Money

PSTAT 5LS – Spring 2026

Q1: Turning Up the Heat

Recall several weeks ago we sent out a PSTAT 5LS Student Survey for you to complete. One of the questions we asked in that Survey was: “Do you prefer hotter temperatures or cooler temperatures?” In this first part of today’s Section Activity, we will be working through a start-to-finish hypothesis test involving this data: specifically, we’ll be examining the proportion of all UCSB students that prefer hotter temperatures over colder temperatures.

Before we dive into our hypothesis testing, it’s important we take a step back and think about the **scope of inference** (remember this from Slide Set 2?)

Q1.1 What is the population of interest? What about the parameter of interest?

- The population is _____
- The parameter is _____

Q1.2 How generalizable do we think the results from this survey are to the population of interest?

- ☐ Quite generalizable
- ☐ Somewhat generalizable
- ☐ Not generalizable at all

Regardless of how we answered the previous question, let’s proceed with our hypothesis testing. Our next steps are to set the null and alternative hypotheses, and select a significance level. Remember that it is very important we do these BEFORE seeing any data!

Q1.3 What should be our null and alternative hypotheses? Make sure you define the parameter in the context of the problem. Additionally, your TA will brainstorm with everyone to figure out what the appropriate null value for this test should be. Use proper mathematical notation!

- The null hypothesis is _____
- The alternative hypothesis is _____

Q1.4 The significance level we'll choose is _____

Here is our data: among the 314 students who filled out the survey, 191 reported preferring hotter temperatures to colder temperatures.

Q1.5 The value of the point estimate is _____ (make sure to use proper mathematical notation!)

Eventually, we'd like to conduct a theory-based hypothesis test of the hypotheses we stated in Question 1.3 above. This means that our next step is to check the conditions needed to invoke the CLT.

Q1.6 Check the two relevant conditions.

Independence:

Success-Failure Conditions:

Regardless of whether or not we believe the conditions are satisfied, we will proceed with our theory-based inference since we want to do a complete example together in section.

Q1.7 The associated p -value is _____ (your TA will walk you through the calculations and give you the final answer).

Q1.8 Formulate a conclusion. Remember that your answer should include:

- Your decision to reject or fail to reject the null hypothesis (and why)
- Whether or not the results are statistically significant
- A clear interpretation of the result in the context of the study

Q2: Donating Money

One of the other questions we asked in the survey was which of several charities you'd donate \$1,000 to (if you had that kind of money). If you recall, you were allowed to select as many of the 7 charity categories provided. In this part of the Section Activity, we'll explore the proportion of students that selected more than one charity to donate their (hypothetical) money to. Of our sample of 311 respondents (removing some non-responses), 228 selected more than one charity. We will use this to construct a confidence interval for the true proportion of UCSB students that would have selected more than one charity to donate their money to (setting aside questions of generalizability for now).

Q2.1 What confidence level should we use? _____ (Your TA will brainstorm with everyone to figure out a preferred confidence level)

Q2.2 What is the center of the confidence interval? _____

Q2.3 Why is this the center of the confidence interval?

Q2.4 Check the success-failure conditions. Make sure to check the right ones!

Q2.5 What is the critical value? _____

Q2.6 Calculate the standard error.

Q2.7 Construct the desired confidence interval, and provide an interpretation.

Q2.8 Does the interval provide evidence that at least 50% of UCSB students would select more than one charity to donate their money to? Explain briefly.